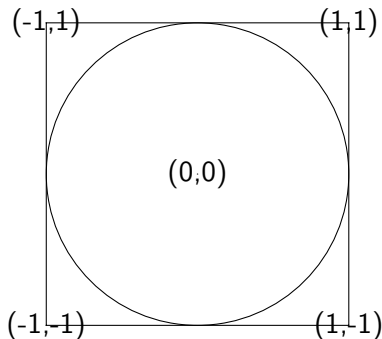


Assignments D

Assignment D (C/C++)

- ▶ Due: April 7 by 2 PM.
- ▶ A simple Monte Carlo simulation can be used to approximate the value of π .
- ▶ The figure below shows a circle with radius $r = 1$ inscribed within a square.



- The area of the circle is $\pi r^2 = \pi$, and the area of the square is $(2r)^2 = 4$. The ratio of the area of the circle to the area of the square, ρ , is given by:

$$\rho = \frac{\text{Area of Circle}}{\text{Area of Square}} = \frac{\pi r^2}{(2r)^2} = \frac{\pi}{4}$$

- ▶ Calculating the value of π using Monte Carlo involves the following steps:
 1. Assume the circle is centered at coordinates (0, 0).
 2. Generate N random points with coordinates (x, y) where x and y are independently drawn from a **uniform distribution** over the interval [-1, 1].
 3. Determine if each point lies inside the circle or not.
- ▶ The value of π can be estimated using the simulation results as follows.
$$\rho = \frac{M}{N} = \frac{\pi}{4}$$
where, N = total number of points generated and M = number of random points inside the circle.
- ▶ Write a program to calculate the value of π using information above, for N = 100, 1000 and 10000. Write the results to the standard output.
- ▶ Vectorizing/parallelizing code is NOT required for this assignment.